

UMA IYER

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EDUCATION

University of Michigan

Aug. 2022 - May 2026

Majors: Data Science and Economics GPA: 3.5

Ann Arbor, MI

Courses: Data Mining, Machine Learning, Deep Learning, Database Management, Economic Forecasting, Advanced

Statistics, Econometrics, Game Theory

Technical Languages/Tools: Python, C++, SQL, PostgreSQL, Spark, PyTorch, NumPy, DuckDB, scikit-learn, Pytest, Poetry, Tableau, Linux, Git, Docker, Excel

London School of Economics Summer Program

Jun. 2024 - Aug. 2024

Courses: International Trade/Finance, Behavioral Economics

London, England

WORK EXPERIENCE

The MITRE Corporation | Data Science Intern

Jun. 2025 - Aug. 2025 | McLean, Virginia

- Developed a **Python package** to automate efficient reading and writing of large geospatial datasets in **PostgreSQL**, supporting batch inserts, SQL query execution, and automatic database connection handling.
- Achieved 98% unit test coverage with **pytest**, created isolated tests to preserve database integrity, ensured reproducibility and dependency management using **Poetry**, and implemented CI workflows via **GitLab CI**.
- Containerized the package with **Docker** and published it to a private registry; now used across MITRE engineers to manage database input/output and standardize I/O operations.

Farcast Biosciences | Machine Learning Intern

Nov. 2024 - Mar. 2025 | Remote

- Built a centralized relational database using **SQL** to unify disjointed lab and research datasets, standardize hematology time-series records to enable consistent data entry and efficient querying.
- Standardized and formatted patient datasets using **Python** and **pandas**, performing cleaning, normalization, and feature engineering on explant and arm-level hematology data to improve consistency and model interpretability.
- Developed **predictive models** using **logistic regression**, **random forests**, and **SVMs** for classification tasks to assess treatment efficacy in cancer patients; applied **statistical techniques** including cross-validation, ROC analysis, and confusion matrix evaluation.

PROJECTS

New York Street Safety Project

Feb. 2026

- Built an **XGBoost model** to identify high-risk NYC intersections and predict primary collision causes.
- Built a **BERT-based** engine to map crash factors to infrastructure fixes via embedding similarities, utilizing **cosine similarity** to retrieve cost-efficient safety interventions for the DOT.
- Developed interactive **ArcGIS** dashboards to perform **spatial-temporal analysis** on 10+ years of collision data and identify high-risk intersections and boroughs.

Data Mining and Frequent Pattern Analysis

Jan. 2025

- Analyzed Movie ratings data using **PySpark** and **FP-Growth** to identify frequently co-watched movie pairs/triplets, generating high-confidence association rules to inform recommendations.
- Processed Amazon review data (120M records) with **Spark DataFrames**, applying frequent itemset mining to identify product co-purchase patterns and recommend items based on past customer purchases.

Predicting ICU Patient Survival

Sept. 2024 - Oct. 2024

- Developed a logistic regression model using **scikit-learn** to predict ICU patient mortality from time-series medical data, performing feature extraction and data preprocessing with **Python** and **pandas**.
- Performed hyperparameter tuning via 5-fold cross-validation and compared linear versus kernel ridge regression models to optimize performance metrics including accuracy and AUROC, visualizing results using **matplotlib**.

LEADERSHIP EXPERIENCE

Michigan Data Science Team | VP Finance, Social Media Manager

Jan. 2024 - Jan 2026

- Acted as an authorized signer, raised over \$5,000 from sponsors, and supervise budget for a 300+ member club.
- Doubled our social media following using campaigns to promote the club and attract sponsors.

ECAS Center at UofM | Tutor

Aug. 2024 - May 2026

- Provide assistance to students with their Programming and Calculus course work.